

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer working on alignment evaluation: finding how and why deployed agentic systems fail, from a systematic risk-downplaying bias in a GenAI tool serving \$200B+ AUM to recipe-induced overfitting in published VLA checkpoints. Also design RL reward signals built to resist gaming, and open-sourced a flow-matching VLA integration for the RLinf training stack. Co-founder of Merlyn Labs; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

- Alignment Evaluation
- Adversarial Evals
- Reward Modeling
- LLM Agents
- Reinforcement Learning
- PyTorch & JAX

WORK EXPERIENCE

BMO AI Centre of Excellence

Sep 2025 - Present

AI Research Engineer | Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management.
- Built the **deterministic agent eval pipeline** that produced the finding: hundreds of synthesized inputs detecting misalignment invisible at anecdote scale.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.
- Developing **RL environments** to train specialized agents for BMO's wealth management division.
- Building internal infrastructure that provisions LLM agents from a written role and scope definition.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.

Merlyn Labs

Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Developing **VLM judges** that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Showed **$\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting**; proposed an alternative baseline.
- Conservative finetuning doubled position-swap success (**21% to 42%**), showing spatial generalization over memorization.
- Open-sourced a **flow-matching VLA integration for RLinf**, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.

Epineuron Technologies

May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop **PeriPulse**, an FDA Breakthrough-designated device now in multi-national clinical trials; designed and assembled PCBs and authored IEC 60601-1 / ISO 13485 validation protocols.
- Optimized power draw via oscilloscope and power-analyzer testing, achieving **900% battery life improvement**.

PROJECTS

Stanford BEHAVIOR-1K Challenge

8th Place, Standard Track

Identified **proprioceptive collapse** as a critical VLA failure mode: masking 60% of proprioception improved task success by up to 48%. Trained manipulation policies on 10,000+ demonstrations covering 22 of 50 scored task types; found chunked execution outperformed temporal ensembling **3x**, exposing missing temporal awareness. Published a [technical report](#) and a [LessWrong analysis](#) of the failure modes.

BardSong

Closed Alpha

Fine-tuned an LLM on D&D sourcebooks for narrative coherence, in closed alpha with **23** dungeon masters. Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video.

EDUCATION

M.Eng in AI & Robotics, University of Toronto

Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University

Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Evaluation & Alignment: Agent Evaluation, Alignment Evaluation, Reward Modeling, LLM Fine-tuning, Reinforcement Learning, Imitation Learning

Systems & Languages: Python, C/C++, PyTorch, JAX, VLA Policies, Flow Matching, OmniGibson, MuJoCo, Sim-to-Real, ROS2